



EETC: An extended encrypted traffic classification algorithm based on variant resnet network

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ABSTRACT

With the advent of big data, encrypted traffic is widely used, and it is gradually becoming a new challenge for network security and management. Many researchers have obtained good results by converting encrypted traffic into images and feeding them to deep learning-based classification models. However, these methods have some limitations in that they cannot do sustainable learning. They have to retrain a new classifier when new traffic is encountered. To tackle this issue, we propose an extended encrypted traffic classification algorithm based on incremental learning. This approach extracts content and statistical information from encrypted traffic and supports multi-label prediction for VPN channels and applications. We can add new classes to the model without completely retraining and accelerate the update cycle of the model. The results show that the proposed model performs well in classification experiments, which can achieve 98.1% and 96.0% classification accuracy on ISCVPN2016 and self-collected dataset respectively. In addition, our method can retain high accuracy under the situation of limited memory space while the number of new classes of data increases gradually. It demonstrates the superiority of constructing a generic forgetting basis for classifying encrypted communication.

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1. Introduction

Due to the prevalence of Internet applications, a large amount of network traffic is generated when people surf the Internet. Therefore, the high-accuracy classification of traffic has become an essential aspect of network security and cyberspace management. For example, the enterprise network administrator can prioritize the quality of service (QoS) for certain specific applications and assign a lower network communication priority to the applications that do not comply with the network policy. Nowadays, the need for information privacy is increasingly recognized. Many companies use encryption protocols that are designed to guarantee the security of information in their products, such as Transport Layer Security (TLS), Virtual Private Network (VPN), Secure Sockets Layer (SSL), Secure Real-time Transport Protocol (SRTP), and so on. However, cyber-attacks are commonplace. Typically, attackers utilize encryption technology to conceal themselves, elude the scrutiny of network security technologies, and cause enormous losses to in-

dividuals and businesses. Therefore, academia and industry have begun to pay attention to the classification of encrypted traffic in recent years.

Traditionally, there are two main methods of traffic classification: the port-based approach and the payload-based approach. In the port-based approach, protocols can be identified directly according to the corresponding port in RFC. For instance, port 21 corresponds to File Transfer Protocol (FTP), whereas port 80 corresponds to Hypertext Transfer Protocol (HTTP). The port-based approach is straightforward. However, with the appearance of dynamically assigned port numbers (Cotton et al., 2011), more and more applications begin to use random ports to circumvent the inspection of the firewall. As a consequence, the port-based approach is invalid. Then, researchers are devoted to the traffic identification method by studying payload in traffic, which is also known as deep packet inspection (DPI) (Bujlow et al., 2015). Due to the specific packet formats in different encryption protocols or applications, such as fixed strings, DPI can explore and summarize the characteristics of payloads from different kinds of traffic. However, it has restrictions with regard to encrypted payload. At present, machine learning has emerged as a strategy to solve the problem of encrypted traffic classification, such as Bayesian technology (Kudryavtsev and Shorgin, 2010), C4.5 decision tree

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(Ho, 1998), support vector machine (Poulet, 2004), and k-nearest neighbor (Sun and Huang, 2010). However, these classical algorithms deeply rely on the artificial feature selection of encrypted traffic. Consequently, deep learning classifiers have been implemented in modern-traffic classification which gives computers the ability to learn the features of traffic. Some works classify encrypted data based on statistical characteristics of traffic. Generally, statistical information needs to be adjusted according to different classification tasks. Apart from that, the majority of these works use raw traffic data to classify. Due to the unified input size of neural network, it is necessary to truncate raw traffic data or add 0 bytes after packet. Hence, the traffic information is lost or redundant, and classification accuracy decreases. In recent years, an increasing number of researchers have utilized parallel network topology to combine the content characteristics and statistical features of encrypted data, which increases the complexity of the network and the training time.

In addition, another difficulty exists in the classification of encrypted traffic. Traditional encrypted traffic classification algorithms often focus on offline training. When new data come, the model needs to be retrained. With the rapid growth of encrypted traffic types, it has become a trend for the dataset to be larger, the model to be more intricate, and the training period to be longer. Therefore, it is of great significance to explore an algorithm that can make the model accumulate learning experience despite limited memory and computational resources.

In this paper, we propose an encrypted traffic classification model based on variant ResNet18 (He et al., 2016) and apply incremental learning to realize expanded classification and improve the model's practicability. Our method is different from the existing approaches of traffic classification in both preprocessing and network structure.

The contributions of our work are as follows:

- We propose a new composition method based on packet splicing to generate three-channel RGB images. It enhances statistical information such as packet direction and data transmission frequency with completely keeping the content information of encrypted traffic.
- To extend the framework, a novel attempt is adopted to apply incremental learning to encrypted traffic classification. We employ a variety of techniques to store a subset of exemplars in limited memory and replay them for the next classification.
- To overcome the bias problem caused by Softmax, we purpose a bias vector and a balance factor to modify the prediction of the model by normalization in an extended framework.
- Our proposed model supports the multi-label prediction of VPN channels and applications. The experimental results show it can achieve 98.1% and 96.0% classification accuracy on ISCXVPN2016 (Draper-Gil et al., 2016) and self-collected dataset respectively.

The remainder of the paper is organized as follows: Section 2 presents related work. Section 3 describes EETC, an extended encrypted traffic classification model based on variant ResNet18 and incremental learning. Section 4 shows the results of experiments, including classification performance and extension performance. In Section 5, we conclude this paper. Table 1 summarizes the acronyms that are used in this paper.

2. Related works

In this section, we review previous related works under two categories: encrypted traffic classification and incremental learning. Table 2 summarizes the input data and network structure using deep learning in encrypted traffic classification.

Table 1
List of acronyms.

Acronym	Expansion
APP	Application
BV	Bias Vector
CIL	Class-Incremental Learning
CNN	Convolutional Neural Network
DER	Dark Experience Replay
DPI	Deep Packet Inspection
EETC	Extended Encrypted Traffic Classification
EWC	Elastic Weight Consolidation
FN	False Negative
FP	False Positive
FTP	File Transfer Protocol
GEM	Gradient Episodic Memory
GIL	General Incremental Learning
HTTP	Hypertext Transfer Protocol
iCaRL	Incremental Classifier and Representation Learning
IP	Internet Protocol
LSTM	Long Short-Term Memory
LWF	Learning without Forgetting
MAC	Media Access Control
NME	Nearest Mean of Exemplars
QoS	Quality of Service
RKHS	Reproducing Kernel Hilbert Space
SAE	Stacked Autoencoders
SRTP	Secure Real-time Transport Protocol
SSL	Secure Sockets Layer
SSR	Shadowsocks-R
TCP	Transmission Control Protocol
TLS	Transport Layer Security
TN	True Negative
TP	True Positive
VPN	Virtual Private Network

2.1. Encrypted traffic classification

In the literature, deep learning-based traffic classification methods are classified into two categories. Some methods directly use the traffic payload as the input to the network. Other methods extract statistical characteristics from the raw traffic as input to the classification model.

Many studies focus on traffic payload. Wang et al. (2017) transformed traffic payloads into images and used an end-to-end one-dimensional convolutional neural network (CNN) architecture to classify these images. This is the first work to turn traffic classification into image classification. They integrated feature extraction and classification network into one system directly using the first 784 bytes of original flow, which can successfully distinguish VPN and non-VPN traffic. He and Li (2020) also proposed an image-based traffic classification method. They converted the first few non-zero payloads of the session into gray images and employed CNN to achieve classification. Hu et al. (2021) proposed the CLD-Net model which can automatically extract features from the packet payload of the original stream and add the information of the packet interval and fuse it into new features. Cui et al. (2019) fed the 28×28 traffic matrices into the capsule neural network (Rajasegaran et al., 2019). Compared with CNN, the capsule network adopts vector instead of scalar as neuron, which can learn the features of packets more effectively. Lotfollahi et al. (2020) proposed a framework called Deep Packet, which preprocessed raw traffic at the packet-level and employed 1DCNN and stacked autoencoders (SAE) (Gehring et al., 2013) in two tasks: application identification and traffic characterization. To cope with packet features, Zou et al. (2018) retrieved flow time-sequence features using Long Short-Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997) paired with CNN. The experimental results showed that their model performed better than using CNN or LSTM alone on encrypted traffic classification.

Table 2
Related works based on deep learning: B(Bytes).

Input Data	Network Structure	Paper
Payload[784B]	1DCNN/2DCNN	Wang et al. (2017)
Payload[784B]	1DCNN	He and Li (2020)
Payload[256B]	1DCNN+LSTM	Hu et al. (2021)
Payload[28×28B]	Capsule Network	Cui et al. (2019)
Payload[1500B]	1DCNN/SAE	Lotfollahi et al. (2020)
Payload[28×28B][3 packets]	CNN+LSTM	Zou et al. (2018)
Payload[784B]	1DCNN+LSTM	Lu et al. (2021)
Payload[1480B]	1DCNN	Soleymanpour et al. (2021)
Arrival time, Frame length, Packet size	2DCNN	Pathmaperuma et al. (2022)
6 fields [20 packets]	LSTM+2DCNN	Aceto et al. (2018, 2019)
Flow-based statistics	CNN	Chen et al. (2017)
Packet direction	1DCNN	Sirinam et al. (2018)
Arrival time, Packet length	2DCNN	Shapira and Shavitt (2019)
IP packet length	BiGRU	Liu et al. (2019)
Payload[784B]+4 fields [32packets]	1DCNN+BiGRU	Aceto et al. (2021)
Payload[64B]+Flow-based statistics	Neural Network	Zhou et al. (2020)
Packet length, Packet direction	Graph Neural Network	Shen et al. (2021)
Payload[32×32×3B]	Variant ResNet18	This Paper

Lu et al. (2021) used the inception module to extract local features of the traffic and the LSTM model to extract the temporal features of the packets within a session. Their approach enhanced the characterization of packet features to improve classification accuracy. Soleymanpour et al. (2021) proposed a Cost-Sensitive CNN (CSCNN) to deal with the class imbalance issue in encrypted traffic classification. It greatly improved the impact of data imbalance on traffic classification. Pathmaperuma et al. (2022) proposed a method to both identify known and unknown traffic and categorize in-app activity types. After filtering the unknown traffic, the classification accuracy of their model would be 92%.

Different classes of encrypted traffic can be identified by evaluating the temporal and spatial information of the flow's packets in light of statistical features. Aceto et al. (2018, 2019) attempted to classify mobile traffic by constructing a classifier based on six statistical characteristics: the number of bytes in the transport layer payload, the TCP window size, the source port, the destination port, the inter-arrival time, and the packet direction. Chen et al. (2017) put forward an online traffic classification algorithm which was called Seq2Img focusing on packet size, inter-arrival time, and direction. They used Reproducing Kernel Hilbert Space (RKHS) embeddings (Rosipal and Trejo, 2001) to transform the packet sequences into six-channel images and classify these images into different applications. Sirinam et al. (2018) applied CNN to website traffic identification. They achieved high accuracy using only the packet direction as an input to the network. Shapira and Shavitt (2019) constructed a two-dimensional histogram as the input to the CNN. They defined the X-axis as the normalized arrival time and the Y-axis as the packet size. Liu et al. (2019) tapped potential flow-based characteristics via an encoder-decoder structure. Aceto et al. (2021) combined statistic characteristics with content characteristics to achieve multimodal traffic classification, ultimately addressing three classification objectives. There exists plain text during a three-way handshake, including unencrypted communication parameters and the public key. Thus, some researchers consider the entropy distribution to be a property of traffic. Zhou et al. (2020) combined entropy estimation with neural networks in their method, where 64 bytes in the content of TCP protocol are randomly selected to calculate entropy. Shen et al. (2021) constructed traffic interaction graphs based on the length and direction of packets, and then used graph neural networks to implement the identification of decentralized apps.

In contrast to the above methods, we leverage both the content characteristics and statistical characteristics of encrypted traffic through the composition method rather than parallel network

architecture. After preprocessing, the variant residual network is used to extract features automatically and realize classification.

2.2. Incremental learning

In incremental learning, the deep model often leads to catastrophic forgetting (Robins, 1993). The model performs well in the current task, but poorly in previous tasks. There are three main categories of methods for solving this problem.

The regularization-based method (Rannen et al., 2017) involves punishing the parameters and encouraging them to be similar to the parameters of the old task. Kirkpatrick et al. (2017) proposed elastic weight consolidation (EWC) based on synaptic plasticity, which mitigated catastrophic forgetting by limiting the space of weight update. Specifically, it includes a penalty term measured by Fisher information within the loss function. Therefore, the weights closely related to the classifier performance are protected. Zenke et al. (2017) suggested limiting changes to the most relevant synapses in order to reduce catastrophic forgetting, thus controlling the changes in important parameters. LWF (Li and Hoiem, 2017) computed the distillation loss between the teacher model and the student model, using the preceding model's output as the soft label to alleviate forgetting.

The structure-based method modifies the architecture of the network (Mallya et al., 2018; Mallya and Lazebnik, 2018), so as to reduce the disturbance between tasks without changing the objective function. You can freeze some weights in the network and keep the same, or specify a model copy for each task. Some studies decrease to reduce the learning rate of the layer shared with the original task and fine-tune it to avoid drastic changes in parameters. Moreover, models using different nonlinear activation functions, such as ReLU, MaxOut, and Tanh, have been proven to improve the effect of incremental learning, especially in MNIST and emotion analysis tasks (Rosenfeld and Tsotsos, 2018). Yan et al. (2021) proposed a novel architecture, which replicated the entire network and added new branches in order to perform the new task. Nonetheless, it will enhance the complexity of architecture.

The replay-based method (De Lange et al., 2021) stores samples in raw format. While learning new tasks, the samples of previous tasks will be replayed. Rebuffi et al. (2017) proposed iCaRL, which adopted herding algorithm to select exemplars and Nearest Mean of Exemplars (NME) classifier for training. DER (Buzzega et al., 2020) employed reservoir sampling (Dash and Ng, 2006) during the whole training phase and investigated the generalization po-

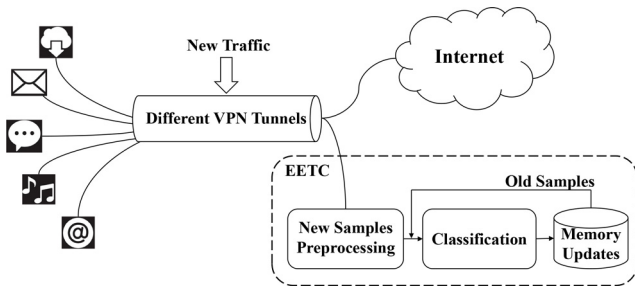


Fig. 1. Overview of EETC with a scenario: EETC is identifying different apps' traffic under different VPN tunnels. When new traffic comes, the model does not need to be trained from scratch through memory updating.

tential of incremental learning. It is an alternative solution to constrain optimization. The key idea of GEM (Lopez-Paz and Ranzato, 2017) is to restrict new tasks without interfering with previous ones.

It is a novel attempt to apply incremental learning to encrypted traffic classification. We refer to the herding algorithm in iCaRL (Rebuffi et al., 2017) and use the variant residual network instead of NME as a classifier. A bias vector and a balance factor are intended to adjust the prediction in order to address the bias problem in softmax. In this paper, we not only evaluate the classification ability of the model but also show the extension performance of the model when new samples are added.

3. Methodology

In this section, we explain the general framework of EETC in detail. The primary goal of EETC is to classify different applications' traffic under different VPN tunnels, as shown in Fig. 1. When new types of traffic occur, EETC's secondary objective is to allow the model to accumulate experience without training from scratch.

Fig. 2 shows the overview of our proposed extended encrypted traffic classification framework, which mainly consists of preprocessing, classification, and memory updating. The raw traffic is used to generate TFRecords files after preprocessing, which are input into neural network for training. Through the Softmax layer, the probability distributions of traffic are created after the automatic extraction of characteristics. In the training phase, we develop an improved cross-distilled loss to mitigate the forgetting of old knowledge. And some rules are formulated to select the old samples after training, where important samples will be stored in memory to prepare for the next classification. Furthermore, our proposed model provides the simultaneous prediction of VPN labels and application labels.

3.1. Preprocessing

The preprocessing of encrypted traffic often varies with the classification task, which can be mainly divided into three levels (e.g., session, flow, and packet). We choose to preprocess at the packet-level in the preprocessing phase due to the need for fine-grained characteristics in application classification. The details of our preprocessing procedure are shown in Fig. 3, consisting of five-tuple clustering, deleting Mac and IP addresses, splicing packets, byte conversion with normalization, RGB images generation, and dividing dataset.

Five-tuple clustering. The five-tuple consists of the source IP address, source port, destination IP address, destination port, and transport layer protocol. Five-tuple clustering can filter irrelevant traffic noise and eliminate the influence of non-target user traffic. For a public dataset, the step of data cleaning is not essential. It is

Table 3

Number of samples in ISCVPN2016.

Class Name	Training Samples	Testing Samples
Chat	6573	731
Email	2586	288
File	25656	2851
P2P	20842	2316
Streaming	26787	2976
VoIP	26700	2967
VPN-Chat	7182	798
VPN-Email	2047	227
VPN-File	25747	2861
VPN-P2P	24324	2703
VPN-Streaming	26678	2964
VPN-VoIP	26603	2956

crucial in self-collected dataset because the original flow contains traffic belonging to some non-target users.

Deleting MAC&IP. In general, the IP address of two hosts will not often change in the process of communication. The MAC address and IP address in each packet are removed to avoid the fixed address feature acquired by the neural network, which leads to model overfitting.

Splicing packets. Different lengths of bytes are padded at the end of each packet. As shown in Fig. 4, we set different padding values (0 or 240) to tackle packets in different directions. We define the packet from the source entity to the target entity as an up-link packet, and its padding value is 0. On the contrary, the packet from the target entity to the source entity is defined as a down-link packet, and its padding value is 240. It is helpful for neural network to learn multimodal features (e.g., content characteristics and statistical characteristics) via different padding bytes between packets.

Byte conversion with normalization. We convert every 3072 bytes (including padding bytes) into a same-size array and normalize each element, which is helpful to accelerate the convergence speed of neural network in the training process.

RGB images generation. Fig. 4 shows how we reshape the normalized array to a 32×32 image with 3 channels. The *packet a* to the *packet n* represents a continuous sequence of packets. Each packet consists of bytes of different lengths. All bytes are converted to a decimal range of 0 to 255. B_{a1} denotes the first byte of *packet a*. Similarly, B_{nk} denotes the *k*th byte of *packet n*. A one-dimensional sequence of 3072 bytes is sequentially reshaped into three 32×32 two-dimensional matrices. The three two-dimensional matrices represent the three channels of the RGB image (R channel, G channel, and B channel). We can construct the RGB image ($32 \times 32 \times 3$) by merging these three two-dimensional matrices. The three-channel RGB images have more information on traffic features than one-dimensional sequences or two-dimensional images.

Dividing dataset. Based on TensorFlow framework (Abadi et al., 2015), the generated three-channel RGB images and labels are converted to TFRecords files which store image data and labels as a sequence of binary strings. The training set and testing set are divided into 9:1. Table 3 shows a detailed description of the number of training and test samples.

3.2. Neural network structure

Residual network is one of the most pioneering works in the field of computer vision and deep learning in the past few years. It effectively solves the problem of vanishing gradient. The neural network structure refers to ResNet18 (He et al., 2016), including four residual blocks with depths of 64, 128, 256, and 512. Each residual block is composed of several residual units, and each

An Extended Encrypted Traffic Classification Algorithm

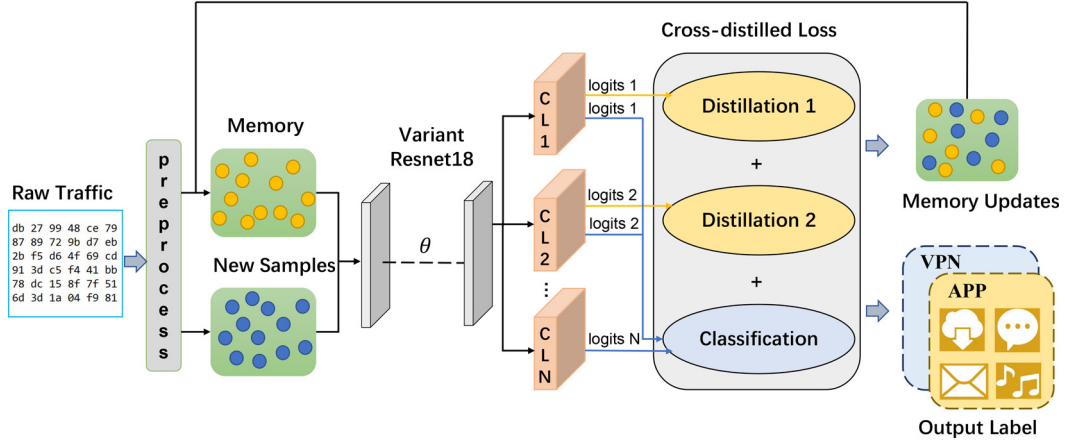


Fig. 2. Overview of EETC: The framework consists of preprocessing, classification, and memory updating. It supports the simultaneous prediction of VPN labels and application labels.

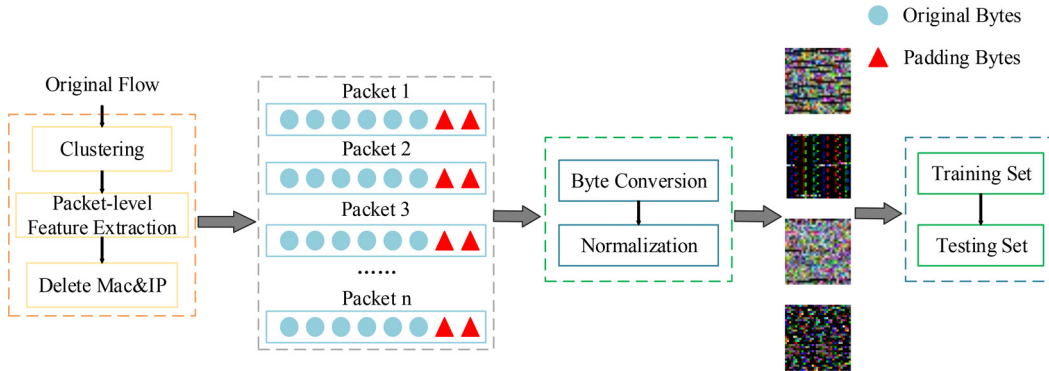


Fig. 3. Preprocessing procedure: padding bytes are in red, original bytes are in blue. The preprocessing completely keeps the content characteristics of the raw traffic without truncating bytes and enhances useful statistical information, such as frequency and direction.

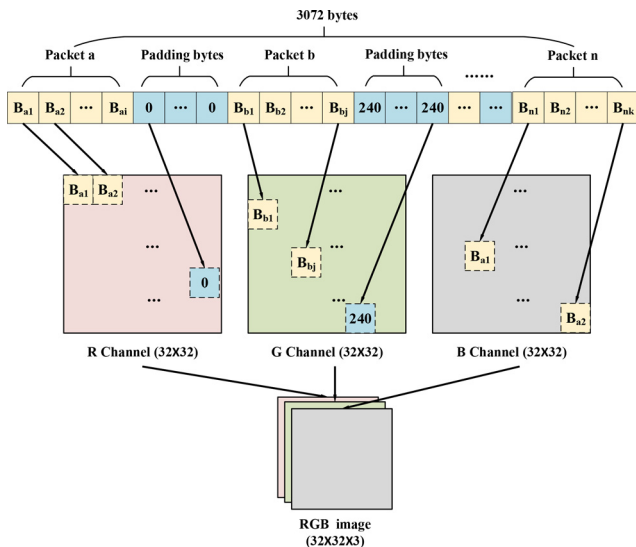


Fig. 4. RGB images generation. The image size of each channel is 32×32 . The padding value of uplink packet is 0 while the padding value of downlink packet is 240.

residual unit uses ($n=6,7,8,9$) convolution kernels with the size of 3×3 .

The classical ResNet18 is commonly used with the default input size of 224×224 . According to the packet length of encrypted traffic, the size of image is designed as 32×32 . Inspired by the phi-

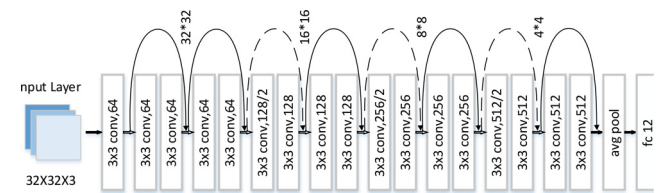


Fig. 5. Variant ResNet18 network structure: The dotted line indicates that there is a skip-connection in the residual block. We replace 3×3 convolution kernels with 7×7 convolution kernels and remove max pooling layer.

losophy of ResNet18, the variant ResNet18 is adopted. In order to keep more fine-grained traffic features, we replace 3×3 convolution kernels with 7×7 convolution kernels and remove the max pooling layer.

The variant ResNet18 network structure is shown in Fig. 5. The entire network starts with an input layer to obtain three-channel RGB images. The convolutional layers have 3×3 filters, extracting the content and statistical information of encrypted traffic. As the core of the residual layer, skip-connection connects the feature maps of different layers to construct the identity mapping. In addition, we perform downsampling to reduce the dimension of the feature map. Finally, the variant ResNet18 ends with a global average pooling layer and a 12-way fully-connected layer with Softmax.

3.3. Loss function and training procedure

In the encrypted traffic classification, we input all class samples in one process. The whole training procedure consists of two

stages: pre-training and fine-tuning. Pre-training aims to preliminarily extract the encrypted traffic characteristics. In addition, we employ the cross-entropy loss for each task:

$$L = -\frac{1}{m} \left[\sum_{i=1}^m \sum_{k=1}^c y_k^i \log(S(x^i)_k) + (1 - y_k^i) \log(1 - S(x^i)_k) \right] \quad (1)$$

where m is the number of training samples, y_k^i represents the class of i th sample. If the i th sample belongs to class k , $y_k^i = 1$. Otherwise, $y_k^i = 0$. $(S(x^i)_k)$ is the output of x^i .

In the variant Resnet18 network, we try to minimize the loss function by AdaGrad (Lydia and Francis, 2019) local optimizer and adopt usual back-propagation for gradient descent.

3.4. Memory updating

Due to the rapid growth and extreme dynamicity of encrypted traffic, it is necessary to provide a way to support continuous model updates (Bovenzi et al., 2021). In this work, we propose an extended traffic classification framework to integrate incremental learning.

The baseline network is the variant ResNet18. Our goal is to meet the following three requirements (Tahir and Loo, 2020):

- It can accommodate the new traffic classes without training from scratch.
- With the growth of classes, the consumption of computer memory and the computing time are increasing slowly.
- The accuracy of classification will not decrease seriously.

It has been demonstrated that replay-based strategies in Class-Incremental Learning (CIL) are simple and effective (Bang et al., 2021). Nonetheless, resolving the disparity between old and new classes is a crucial obstacle for replay-based techniques (Hou et al., 2019). With the increase of training epochs, we adopt representative memory updating to prioritize and select training exemplars. When a new class of traffic or a group of new traffic data is added to the current model, a subset of representative samples will be selected and stored in the representative memory. S denotes the computer's fixed memory capacity for the model. The number of samples per class k is given by $k = S/m$, where m is the number of classes stored in memory.

The updating of representative memory consists of two parts. First, after the training of each incremental stage, we will store the old class samples in a sorted list, keep the first k and remove the others that are no longer needed. Second, we calculate the average Euclidean distance of the new class samples in the current class. The new samples are sorted according to the distance from the current class mean, and the first k samples are preferred for storage.

3.5. Improved cross-distilled loss

As illustrated in Fig. 2, the feature extractor sends features to the classification layer with logit output. During the training phase, the model employs these logits to calculate the gradient of the weight in neural network through cross-distilled loss, which is composed of classification loss and distillation loss. We try to measure the classification performance difference of all classes in terms of classification loss. We will then suppress the difference between the previous model and the current model in distillation loss. Notably, the structure of the feature extractor has not changed. Only the new classification layer is connected to the feature extractor. Paper (Li and Hoiem, 2017) proposed the following loss function in the training model:

$$\text{Loss} = \sum_{x_i, y_i \in G} (1 - \gamma) \times C_{\text{entropy}}(F(x_i), y_i) + T^2 \gamma \times C_{\text{entropy}}(F^T(x_i), H^T(x_i)) \quad (2)$$

where G is training dataset, C_{entropy} is cross-entropy loss function, and y_i is ground truth of new classes. $F(x_i)$ is exactly the same as $F^T(x_i)$, giving a distribution over the old classes and new classes. $H^T(x_i)$ is the model that outputs the probability distribution over old classes. And temperature parameter T to soften the final Softmax layer.

Hinton et al. (2015) found the bias problem that models tend to be more prone to new traffic classes. Mai et al. (2021) showed that Softmax classifier and related fully connected layer are easily affected by the imbalance between old and new classes. Bias problem (Javed and Shafait, 2018) is one of the main sources of catastrophic forgetting. We specifically analyze the causes of bias problem:

- Modification of network structure. The model is required to add weights in fully connected layer when receiving new classes.
- Decoupling of feature extraction and classification. The weight of classifier will be updated with the weight of encoder.

The paper (Javed and Shafait, 2018) proposed that representative memory and knowledge distillation are incapable of solving the bias of neural network in the training process. Motivated by this, we propose an extended algorithm based on bias vector, in which a bias vector V is proposed to represent the statistics of predict labels and real labels. This demonstrates that the bias vector V intuitively reflects the degree to which class is biased by the model. Then, a balance factor α is defined to modify the prediction of the model by normalization.

$$V = \sum_{x_i, y_i \in G} (1 - \gamma) \times y_i + T^2 \gamma \times H^T(x_i) \quad (3)$$

$$\alpha = \frac{\|V\|}{V} \quad (4)$$

Finally, the proposed loss function is as follows:

$$\text{Loss} = \sum_{x_i, y_i \in G} (1 - \gamma) \times C_{\text{entropy}}(\alpha \cdot F(x_i), y_i) + T^2 \gamma \times C_{\text{entropy}}(\alpha \cdot F^T(x_i), H^T(x_i)) \quad (5)$$

Algorithm 1 describes the training process of extended algorithm based on bias vector in detail.

3.6. General incremental learning

General Incremental Learning aims at learning from stream data and does not rely on boundaries between tasks during training and testing. Because the encrypted traffic appears in the form of data stream, the traffic sampling shall be processed in sequence. Sequential sampling has two main challenges: 1) the input stream must be processed in one process and 2) the size of the stream is usually unknown in advance. To solve the above problem, reservoir sampling (Dash and Ng, 2006) is adopted, which selects uniform random samples of a given size from the input stream without replacement.

Herding algorithm (Rebuffi et al., 2017) needs to calculate the sample distance from the mean of current class, which increases the model's computing cost. To explore the generalization capabilities of the algorithm, we substitute the more selective sampling method with reservoir sampling. Without knowing the length of input stream in advance, each traffic sample has an equal chance of being stored in memory S . The reservoir sampling accelerates the algorithm and makes general incremental learning more competitive. Considering the possibility of a sudden distribution shift (Yu et al., 2020) in the input stream, we replay the samples from previous tasks and use two additional coefficients λ and μ to balance. To this end, the loss function is combined with regulariza-

Algorithm 1 Extended Algorithm Based on Bias Vector.

Input $O = (O_1, \dots, O_{m-1})$ //old exemplar set
Input $C = (C_m, \dots, C_t)$ //current exemplar sets
Input S //memory size
Require current feature function $\varphi(x)$
 $G = O \cup C$ //combined training set
network training with loss function

$$\text{Loss} = \sum_{x_i, y_i \in G} (1 - \gamma) \times C_{\text{entropy}}(\alpha \cdot F(x_i), y_i) + T^2 \gamma \times C_{\text{entropy}}(\alpha \cdot F^T(x_i), H^T(x_i))$$

$$V = \sum_{x_i, y_i \in G} (1 - \gamma) \times y_i + T^2 \gamma \times H^T(x_i)$$
 //bias vector

$$\alpha = \frac{\|V\|}{V}$$
 //balance factor
that consists of distillation and classification
 $k \leftarrow S/m$ //number of exemplars per class
for $y = 1, \dots, m - 1$ **do**
 $O \leftarrow (o_1, \dots, o_k)$ //keep only first k
end for
for $y = m, \dots, t$ **do**
 $\mu \leftarrow \frac{1}{n} \sum_{x \in X} \varphi(x)$ //current class mean

$$C_s \leftarrow \arg \min_{x \in X} \left| \mu - \frac{1}{s} \left| \varphi(x) + \sum_{j=1}^{s-1} \varphi(C_j) \right| \right|$$

select samples with the minimum distance from the mean of current class
end for
 $O \leftarrow (O_1, \dots, O_t)$ //new exemplar sets
carry out multi-stage incremental learning training.

tion:

$$\text{Loss} = \sum_{x, y \in S} C_{\text{entropy}}(F(x), y) + \lambda \sum_{x', y', z' \in S} \|z' - h(x')\|_2^2 + \mu \sum_{x'', y'', z'' \in S} C_{\text{entropy}}(F(x''), y'') \quad (6)$$

where $h(x)$ is output logits. $F(x) \triangleq \text{softmax}(h(x))$, giving a distribution over the old classes and new classes. And $z \triangleq h_t(x)$ when t indicates an incremental task.

Algorithm 2 elaborates on the training process of extended algorithm based on general incremental learning.

4. Experiments

This section details the classification performance and extension performance for EETC. We first verify the classification per-

Algorithm 2 Extended Algorithm Based on General.

Incremental Learning

Input: dataset G , coefficients λ and μ , memory buffer S

for (x, y) **in** G **do**

$(x', y', z') \leftarrow \text{sample}(S)$

$(x'', y'', z'') \leftarrow \text{sample}(S)$

$z \leftarrow h_t(x)$

network training with loss function

$$\text{Loss} = \sum_{x, y \in S} C_{\text{entropy}}(F(x), y) + \lambda \sum_{x', y', z' \in S} \|z' - h(x')\|_2^2 + \mu \sum_{x'', y'', z'' \in S} C_{\text{entropy}}(F(x''), y'')$$

$S \leftarrow \text{reservoir sampling}$

end for

Table 4

Classes in three datasets.

Dataset	Classes
ISCXVPN2016	Chat, Email, File, P2P, Streaming, VoIP, VPN-Chat, VPN-Email, VPN-File, VPN-P2P, VPN-Streaming, VPN-VoIP
Self-collected	SSR-YouTube, SSR-Zalo, SSR-Weibo, SSR-Wechat, Psiphon-YouTube, Psiphon-Zalo, Psiphon-Weibo, Psiphon-Wechat
TLS	HTTPS, SMTPS, POP3S, FTPS

formance on two datasets in the offline model. We then set up incremental tasks in a phase-by-phase manner to evaluate the extension performance of the two algorithms by performing a single update that adds different traffic classes. Finally, we show that our algorithm correctly adjusts the bias to the prediction of new traffic classes.

4.1. Dataset

To confirm the validity and practicability of the algorithm, we experiment on three datasets: a public ISCXVPN2016 traffic dataset (Draper-Gil et al., 2016), a self-collected traffic dataset, and a TLS traffic dataset. Table 4 shows the classes in each dataset.

ISCXVPN2016 which is based on OpenVPN channel and captured by Wireshark and TCPDUMP is provided by the Canadian Institute of cyber security. The dataset includes both non-VPN and VPN traffic and involves many popular applications. We refer to DeepPacket (Lotfollahi et al., 2020) to label the raw traffic in the dataset. The labeled ISCX dataset has 12 classes, including 6 classes of VPN traffic and 6 classes of non-VPN traffic.

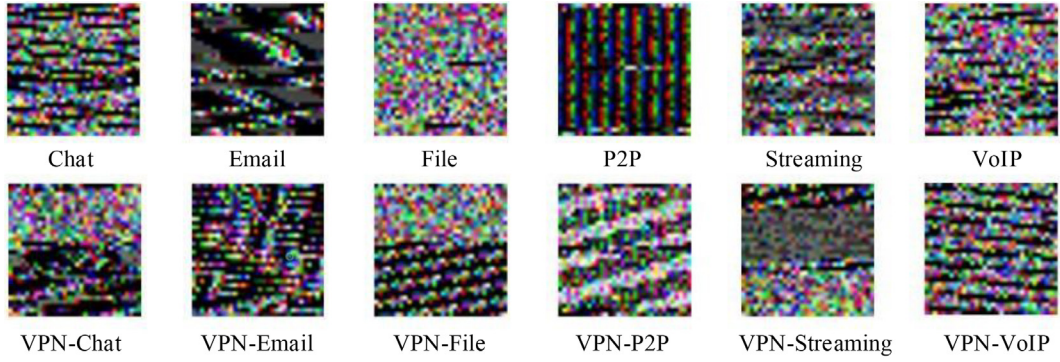
Most public datasets mainly focus on traffic types or application types. To solve the problem, we expand the experimental data by capturing different applications' traffic under different VPN channels. In detail, we choose Wireshark to capture the raw traffic generated by Android phones after running scripts to verify the practical value of the algorithm. The self-collected dataset consists of eight kinds of encrypted traffic samples based on four kinds of APP (e.g., YouTube, Zalo, Weibo, WeChat) under two kinds of VPN channels (e.g., SSR and Psiphon). In the process of running scripts, aiming at ensuring the diversity and authenticity of samples, text, video, voice, static picture, dynamic picture, and other message formats are applied automatically with each message sent at random intervals. In addition, we also collect a TLS-related dataset. This dataset has 4 classes of TLS traffic (HTTPS, SMTPS, POP3S, FTPS).

4.2. Preprocessed results

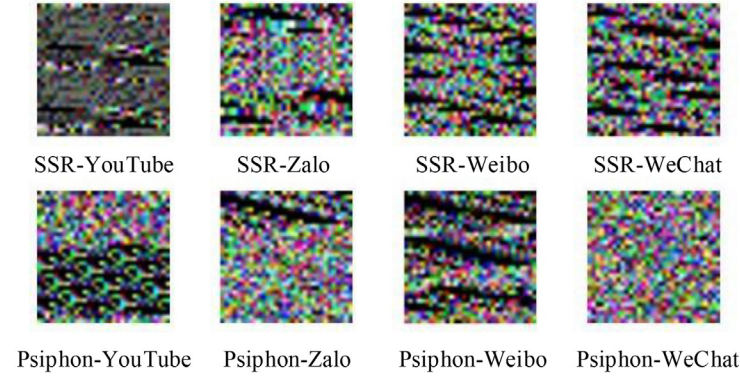
We preprocess the three datasets and then visualize RGB images, as shown in Fig. 6. It is apparent that different traffic classes have different kinds of textures. Therefore, it is reasonable to assume that our method can effectively discriminate traffic features.

4.3. Hyper-parameters selection

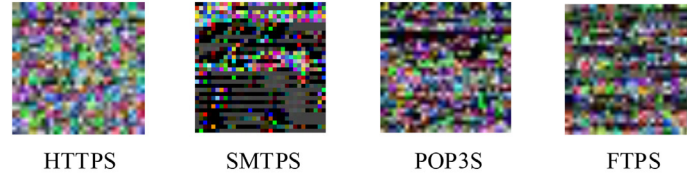
TensorFlow is used as a deep learning framework that runs on Ubuntu 16.04 64bit OS. The specific experimental environment is shown in Table 5. All the experiments on ISCXVPN2016 are run for 50 epochs per increment with a batch size of 64, a momentum of 0.9, a weight decay of 0.00005, a temperature of 2, and a memory budget of 2000. At the beginning of training, the learning rate is set to 2.0 and then reduced 5 times in 45, 60, and 68 epochs. In the improved cross-distilled loss function, γ is set to 0.5.



(a) Visualization of all classes of traffic in ISCXVPN2016 dataset



(b) Visualization of all classes of traffic in self-collected dataset



(c) Visualization of all classes of traffic in TLS dataset

Fig. 6. Visualization of encrypted traffic. We show the generated three-channel RGB images from (a) ISCXVPN2016 dataset, (b) self-collected dataset, and (c) TLS dataset. It is apparent that three-channel RGB images with different traffic classes have unique texture information, which will be conducive to classification.

Table 5
Experimental environment.

Category	Parameter
System	Ubuntu 16.04 64bit OS
CPU	Intel(R) Core i5-8600K
Memory	8GB
Graphics card	NVIDIA GTX 1080Ti
Deep learning backend	TensorFlow1.15
Cuda version	9.0
CuDNN version	7.4

4.4. Competing methods

To demonstrate the effectiveness of EETC in encrypted traffic classification task, we compare the below algorithms under the same condition, which are different in both preprocessing and network structure.

- 1DCNN (Cui et al., 2019): It integrated feature extraction and classification network into one system directly using the first 784 bytes of original flow, which can successfully distinguish VPN and non-VPN traffic.

- SAE (Gehring et al., 2013): It preprocessed raw traffic at packet-level and employed 1DCNN and stacked autoencoders in two tasks (e.g., application identification and traffic characterization).
- CNN+LSTM (Zou et al., 2018): It extracted the flow time-sequence features by Long Short-Term Memory combined with CNN to deal with packet features. To be specific, CNN extracts the content features of a single packet while LSTM extracts the statistical characteristics of any three sequential packets in the flow.

4.5. Classification performance

Four measures are utilized for evaluation: accuracy, precision, recall, and F1. For the multi-classification problem, we regard each class and the remaining classes as binary classification. Eqs. (7)–(10) is shown below, where TP , TN , FP , and FN respectively stand for true positive, true negative, false positive, and false negative.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (7)$$

$$Precision = \frac{TP}{TP+FP} \quad (8)$$

Table 6
Comparison of different lengths of padding bytes (%).

Lengths of padding bytes	Accuracy	Precision	Recall	F1
0	97.7	96.6	95.3	95.7
24	98.1	96.7	95.8	96.1
48	97.3	95.1	95.1	95.0

Table 7
Comparison of different size of RGB images (%).

Size of RGB images	Accuracy	Precision	Recall	F1
32×32×3	98.1	96.7	95.8	96.1
64×64×3	97.2	95.6	95.1	95.3
128×128×3	96.5	93.9	94.9	94.4

Table 8
Comparison of composition of RGB images (%).

Composition	Accuracy	Precision	Recall	F1
Payload	83.7	84.2	83.5	83.7
Header + Payload	98.1	96.7	95.8	96.1
Improvement	14.4	12.5	12.3	12.4

Table 9
Comparison of different network structures (%).

Network	Accuracy	Precision	Recall	F1
ResNet18	90.0	85.5	83.6	84.2
Variant ResNet18	98.1	96.7	95.8	96.1
Improvement	8.1	11.2	12.2	11.9

$$Recall = \frac{TP}{TP+FN} \quad (9)$$

$$F1 = \frac{2TP}{2TP+FP+FN} \quad (10)$$

Three design choices are investigated in EETC: 1) the length of padding bytes, 2) the composition of three-channel RGB image, and 3) the type of classifier.

In preprocessing, we study the importance of padding bytes. Table 6 shows the effect of the length of padding bytes on model evaluation. Depending on the image size, we set the padding length to 0, 24, and 48. As shown in Table 6, the proposed model achieves the highest accuracy of 98.1% when the padding length reaches 24. Our experimental results indicate that the padding bytes with appropriate length are conducive to generating a more discriminative traffic image and enhancing the classification ability of the network.

We study the effect of different sizes of RGB images on the classification results. It can be seen from Table 7, the classification results show a slight decrease as the size of the image increases. It shows that the 32×32 size of the traffic images is more beneficial to the classification results of the network.

To verify whether the packet header affects the effectiveness of the encrypted traffic classification algorithm, under the same condition, we conduct a comparative experiment on the composition of input RGB images. As shown in Table 8, it is indicated that the packet header contains information beneficial to the identification of encrypted traffic, and accuracy is increased by 14.4%. As is common knowledge, a packet header usually contains a series of fixed features, such as packet length, protocol type, live time, version number, etc.

Table 9 reveals the difference between the original ResNet18 network and the Variant ResNet18 network for training. The Vari-

Table 10
Comparison of different algorithms (%).

Algorithm	Accuracy	Recall	F1
1DCNN	85	86	86
SAE	92	92	92
CNN+LSTM	91	91	91
This Paper	98	96	96

ant ResNet18 network has better accuracy under small image input, which gets an improvement of 8.1%.

Compared with the three encrypted traffic classification algorithms below, as shown in Table 10, our approach outperforms existing approaches on ISCVPN2016, achieving the highest accuracy of 98%, recall of 96%, and F1 of 96%.

The confusion matrix is utilized to evaluate the performance of our proposed model. As shown in Fig. 7, we visualize the confusion matrix of the classifier on ISCVPN2016, self-collected dataset, and TLS dataset. It can achieve 98.1%, 96.0%, and 97.3% classification accuracy respectively.

The experimental results show that the accuracy of all traffic classes is higher than 90% except for e-mail (see Fig. 7(a)). The reason is that the size of e-mail pcap files is smaller than other classes in ISCVPN2016, which results in fewer training samples. Therefore, we pay attention to the balance of raw traffic in the self-collected dataset. As a result, each class of self-collected datasets is very accurate. For example, we achieve an accuracy of 99%, 96%, and 100% classifying p-weibo, p-wx p-youtube (see Fig. 7(b)). It demonstrates that the proposed method can successfully identify different apps' traffic in different VPN channels. In addition, we also provide the results for TLS-related traffic. The classification accuracies of four TLS traffic (HTTPS, FTPS, SMTPS, POP3S) are 99%, 98%, 95%, and 97% (see Fig. 7(c)). It indicates that the proposed method can successfully identify TLS-related traffic.

4.6. Extension performance

For the public dataset, we divide 12 traffic classes into 2,3,4, and 6 incremental training steps, that is to say, performing a single update adding 6,4,3, and 2 traffic classes.

To validate the extension performance of EETC, we compare two incremental learning strategies: BV (i.e., Bias Vector) and GIL (i.e., General Incremental Learning) in four scenarios. Taking memory size as a key index, we study the classification performance under different memory sizes. In general, it has better performance in larger incremental steps, because the student model obtains less distillation loss by reducing the noise caused by training. As shown in Fig. 8, the accuracy of the classifier has not decreased rapidly in BV. In the last incremental task, most of them reach more than 85%.

Compared with BV, GIL has the advantage of less performance dropping rate in medium and large memory. We establish a performance dropping rate that measures the absolute drops in the last task with respect to the accuracy in the first task Zhang et al. (2021). We believe that random sampling strategy and improved loss function with regularization are the main reasons for better performance. Meanwhile, it leads to a minor disadvantage that the accuracy of the initial task is lower than BV. However, extension performance focuses more on the degree of forgetting. To sum up, when the memory buffer is small, BV shows the best performance in ISCVPN2016. With a larger memory size, GIL suffers from less forgetting than BV.

Furthermore, we attempt to determine whether the bias vector and balance factor play a role in the CIL. Fig. 9 shows the confusion matrix of BV in the last incremental task, from which we can find that in the process of constantly adding new classes to training

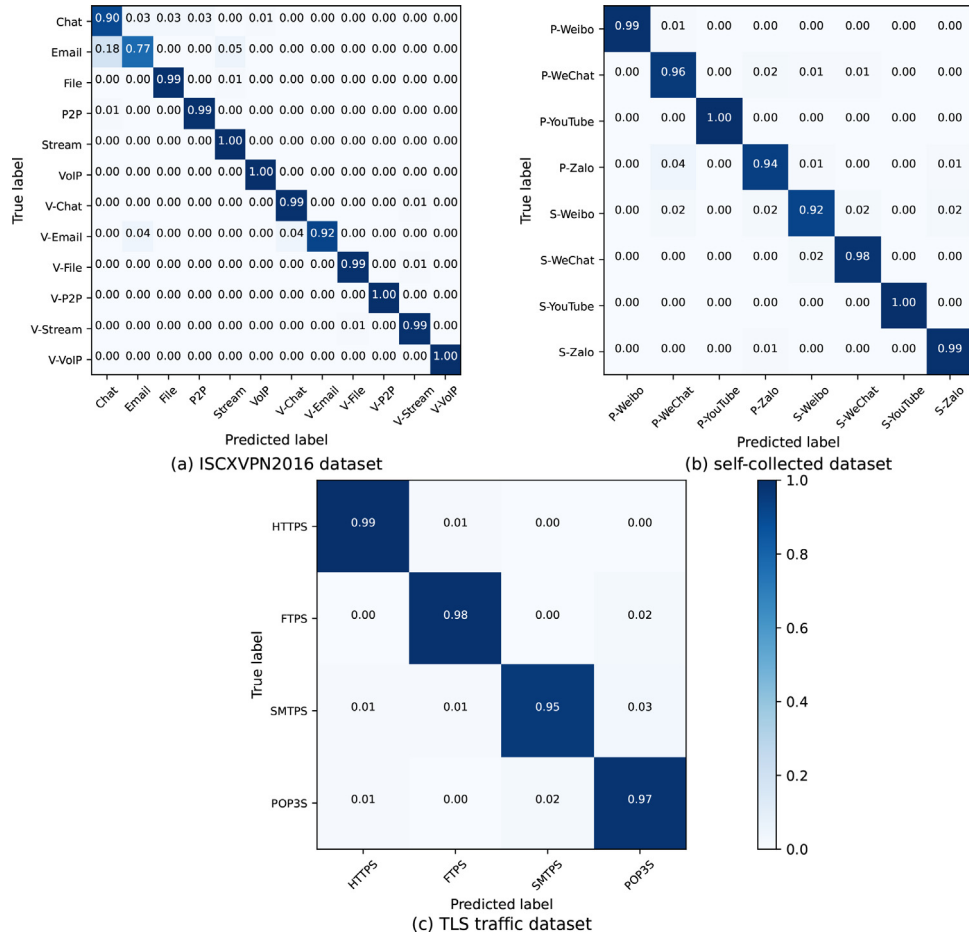


Fig. 7. Confusion matrix on different datasets. The value on the diagonal represents the classification performance. The higher the value, the better the classification performance.

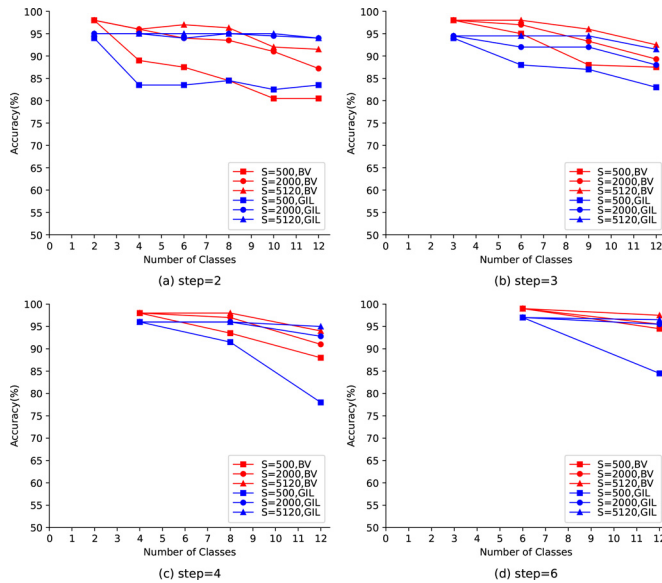


Fig. 8. Classification accuracy when performing increments of different sizes with BV and GIL.

samples, the classifier does not forget the old traffic classes and our model successfully overcome catastrophic forgetting in incremental learning.

Table 11

Time cost of different algorithms.

Algorithm	Training time	Testing time
1DCNN	50.5 min	0.09 s
SAE	47.2 min	0.08 s
CNN+LSTM	75.6 min	0.13 s
BV	26 min	0.10 s
GIL	34 min	0.10 s

Additionally, we present the performance of GIL under different hyper-parameters when incremental step=2, as shown in Fig. 10. To the end, λ is set to 0.2 and μ is set to 0.5 in GIL.

Table 11 shows the time cost of several algorithms. The training time of our proposed BV and GIL for a single incremental task is significantly less than that of other algorithms. It is due to other algorithms do not have the ability to extend, and the entire network needs to be retrained for each incremental task. Among them, BV has the shortest average training time of 26 min. This is because the teacher model in BV only calculates all probability labels at the beginning of each incremental stage. Regarding the teacher model as the student model of the previous incremental task, its parameters have been fully updated. As a result, the parameters in this incremental stage are frozen and only used to generate the probability labels of data. The teacher model without updating parameters does not need to predict the same data many times. Due to the trick, BV occupies a relatively large speed advantage in the training phase. All algorithms are based on deep learning, so the test

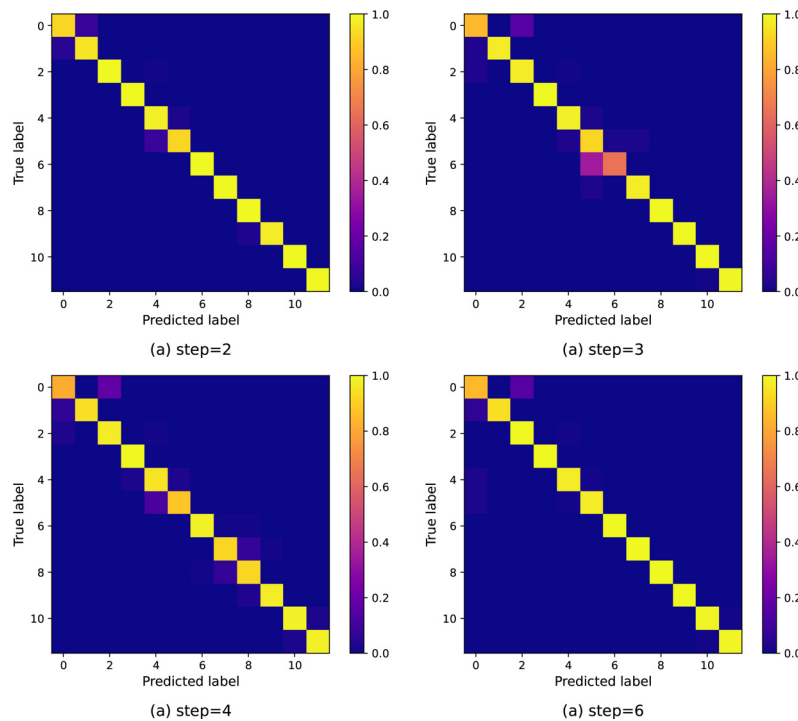


Fig. 9. Confusion matrix of BV on different incremental steps.

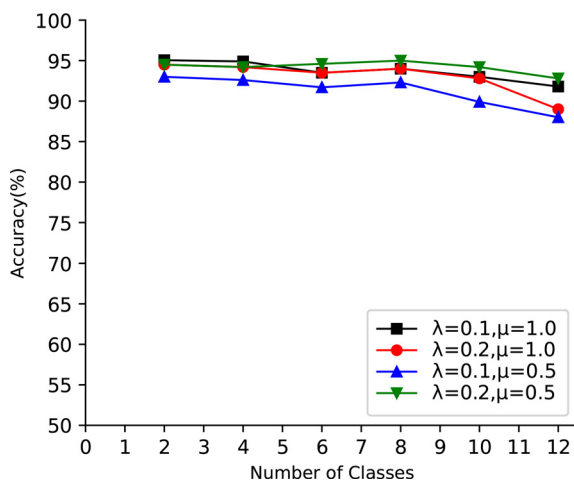


Fig. 10. Performance of GIL under different hyper-parameters with step=2.

times are roughly the same. 1DCNN and SAE have fewer network layers, so the test times are slightly lower than the other methods.

In a word, the two methods have their own advantages. It can be concluded that our methods show the superiority of building a general unforgetting base for encrypted traffic classification.

5. Conclusion

In this paper, we propose an effective extended encrypted traffic classification model based on the variant ResNet18. Notably, the multi-channel images are composed of raw traffic with padding bytes. It perfectly combines content characteristics and statistical characteristics. As a result, different apps' traffic can be identified in different VPN channels. To support continuous model updates, incremental learning is applied to achieve extended classification, which is a novel attempt at encrypted traffic classification. Our method not only shows significant superiority over other methods

in the public dataset but also has stable performance in the self-collected dataset, which is composed of the traffic of popular apps in the real world. When new classes of traffic are added continuously, our methods can maintain high accuracy under the condition of limited memory space without catastrophic forgetting. We propose a method to establish a generic unforgetting basis for encrypted traffic classification. In the future, we will focus on the following: 1) designing a special classifier for encrypted traffic classification instead of a general deep learning classifier, 2) publishing our self-collected traffic dataset, and 3) integrating other mechanisms to boost extension performance.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Rui Wang reports financial support was provided by National Natural Science Foundation of China.

CRediT authorship contribution statement

Xiuli Ma: Conceptualization, Methodology, Visualization, Writing – review & editing. **Wenbin Zhu:** Methodology, Data curation, Writing – original draft. **Jieling Wei:** Methodology, Data curation, Writing – review & editing. **Yanliang Jin:** Conceptualization, Methodology, Visualization. **Dongsheng Gu:** Supervision, Writing – review & editing. **Rui Wang:** Supervision, Writing – review & editing.

Data availability

Data will be made available on request.

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